

Elder Tracking and Fall Detection System using Smart Tiles

M. Daher*, A. Diab, M. El Badaoui El Najjar, M. Khalil and F. Charpillet

1

Abstract— Fall detection for elderly and patient is a very important service that has the potential of increasing autonomy of elders while minimizing the risks of living alone. It has been an active research topic due to the fact that health care industry has a big demand for products and technology of fall detection systems. Owing to the recent rapid advancement in sensing and wireless communication technologies, fall detection systems have become possible. They allow detecting fall events for the elderly, monitoring them, and consequently providing necessary help whenever needed. This paper describes the ongoing work of detecting falls in independent living senior apartments using force sensors and 3-axis accelerometers concealed under intelligent tiles. The force sensors permit detecting elders' falls, locating, tracking and recognizing human activities (walking, standing, sitting, lying down, falling, and the transitions between them). However, the detection accuracy on real data contains false alarms coming from falling and lying postures. To solve this issue, we propose the fusion between the force sensor measurements and the accelerometer sensor decisions. As a consequence, the system accuracy is satisfactory and the results show that the proposed methods are efficient, and they can be easily used in a real elder tracking and fall detection system.

Index Terms— Elderly fall detection; parameters selection; postures; sensing floor; signal detection and processing.

I. INTRODUCTION

NOWADAYS, aging population shows a drastic increasing during the past decade; it's one of the greatest social and economic challenges of the 21st century [1]. Advances in medicine and public health services have improved the overall longevity of people, and it is expected that over the next 50 years, the proportion of people aged more than 60 years is expected to be doubled from 10% to 22% [2]. Those people prefer growing old at home and saving their independent lifestyles that often come with high risks. According to the statistics of Centers for Disease Control and Prevention (CDC), one out of three adults age 65 and older falls each year in the United States, and 61% of these falls occur at homes that cause

10,000 deaths. Yet obtaining a quick assistance after a fall reduces the risk of hospitalization by 26% and the death by 80% [3]. Accordingly, many supportive technologies and systems have been developed to track and monitor activities of elderly persons at home in order to assist their independent living and reduce the cost of premature institutionalization. But generally, all of these systems are relying on only one data provider (movement-sensor, camera, or accelerometer, etc.) that have their own limitations and do not ensure 100% reliability [4]. Moreover, there still is a lack in experience and systematic knowledge to intelligently assemble the components into a robust, friendly-user and effective system, making no false alarm and detecting each fall case, without affecting elderly daily living patterns [5].

This paper describes an approach of intelligent fall detection system using multi data providers that aim to provide more accuracy, precision and efficiency. Yet our main source of data is the ground sensors network within the INRIA-Nancy platform [6,7].

II. FALL DETECTION SYSTEM

There have been a lot of fall detection techniques proposed since the early 1990s. It can be said that most fall detection systems runs on the same mechanism. "Fig. 1" shows the general framework of a fall detection system [8].

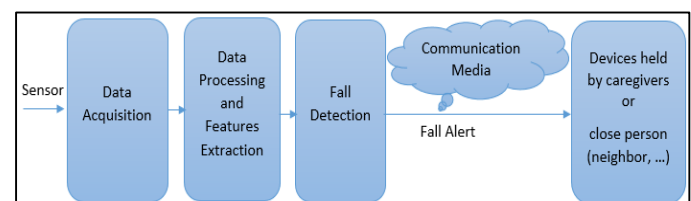


Fig. 1. General framework of a fall detection system.

Fall can be detected using wearable sensor, vision based, and ambient sensor. All three techniques have own advantages and disadvantages based on accuracy, time efficiency, privacy, etc. We choose working on ambient based fall detection using

Manuscript submitted on August 23, 2016. This work is supported by the Lille University - Science and Technology, the INRIA Research Center in Nancy Grand Est, the Lebanese University – Azm Center for Biotechnology Research, and the Université de Technologie et de Sciences Appliquées Libano-Française (ULF).

*M. Daher is with the Lille University - Science and Technology, CRISTAL, UMR 9189, 59650 Villeneuve d'Ascq, France (e-mail: mohamad.daher@univ-lille1.fr)

A. Diab is with Lebanese University, Azm research center in biotechnology, EDST, Lebanon (e-mail: ahmaddiab@ul.edu.lb).

M. El Badaoui El Najjar is with the Lille University - Science and Technology, CRISTAL, UMR 9189, 59650 Villeneuve d'Ascq, France (e-mail: maan.el-badaoui-el-najjar@univ-lille1.fr).

F. Charpillet is with Inria, Villers-lès-Nancy, F-54600, France, also with CNRS, Loria, UMR n° 7503, Vandœuvre-lès-Nancy, F-54500, France, and also with Université de Lorraine, Loria, UMR n° 7503, Vandœuvre-lès-Nancy, F-54500, France (e-mail: françois.charpillet@inria.fr).

M. Khalil is with Lebanese University, Azm research center in biotechnology, EDST, also with CRSI research center, Lebanon (e-mail: mohamad.khalil @ul.edu.lb).

sensing floors to the non-intrusive advantage for elder due to their invisibility, as well as both in terms of user privacy and tile perception. So elder can get profit of his normal life and a hidden monitoring system.

A. State of the art

Sensing floors are floors equipped with sensors for various practical applications. The use of floor-sensors in ambient intelligence contexts began in the late 1990's, with projects like the ORL active floor [9] by Addlesee et al., the Magic carpet [10] by Paradiso et al., and the Smart floor [11] by Orr and Abowd, where they provided information for reasoning about the observed space. These floors were later on integrated into smart environments, aimed at delivering assistance services like continuous diagnosis of users' health. These smart environments also integrated assistive robotic technologies with sensing networks. Examples of such environments include the Gator Tech Smart House made by the University of Florida [12], the Aware Home introduced by the Georgia Institute of Technology [13], the Robotic Room system [14, 15] developed by the University of Tokyo, and the ActiSen activity-aware sensor network [16] jointly developed by researchers from several US universities. Further advances were brought by research on human footprint biometrics [17, 18, 19], which allowed sensing floors to recognize humans by extracting features from footprints and human gait. Also Braun et al. introduced a floor with capacitive sensors, called CapFloor [20], that can localize humans using low-intensity electric fields, and which was used for fall detection. Ropponen et al. used a floor with a low-frequency RFID location system, based on a matrix of quad antennae under the floor surface [21]. This floor was used for tracking people [22], and for detecting human falls [23]. The Institute for Biomedical Technology and Technical Medicine (MIRA), University of Twente-Netherlands, proposes a method to assess foot placement during walking using an ambulatory measurement system consisting of orthopedic sandals equipped with force/moment sensors and inertial sensors [24]. Moreover, The Department of Intelligent Mechanical Systems Engineering, Kochi University of Technology, presents a wearable ground reaction force (GRF) sensor system and its application to measure the extrinsic gait variability [25]. In 2014, T. Liu et al. developed a wireless sensor system for lower limb dynamics analysis using wearable three-dimensional gait analysis system [26]. In 2015, D. Li et al. introduced an inductive sensor for real time measurement of plantar normal and shear forces distribution on diabetes patient's foot that can provide useful information for physicians and diabetes patients to take actions in preventing foot ulceration [27]. Finally, some systems are already commercially available today, like the SensFloor [28,29] (a floor network of capacitive proximity sensors), and FloorInMotion [30].

We have surveyed the literature on ambient intelligence systems. We will introduce now the sensing floor prototype developed by LARSEN Team at INRIA Nancy – Grand Est, which will be used throughout this paper.

B. The INRIA-Nancy Smart Tiles Prototype

The INRIA-Nancy sensing floor consists of 104 tiles (60*60 cm). Each tile is equipped with a 3-axis accelerometer in the center of the tile, and four force sensors positioned at each corner [31,32]. Tiles support a real-time process which ensures communication with their neighbors and any agent laid on them using ZigBee wireless technology. A schematic view of this apartment composed of smart tiles with their identifiers is showed in "Fig. 2".

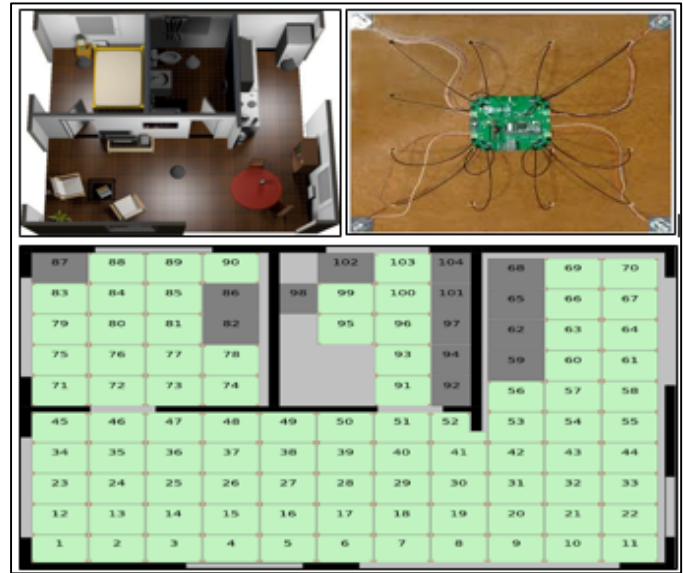


Fig. 2. At the top-left is the INRIA-Nancy smart apartment, at the top-right is the underside of a smart tile, and at the bottom is the schematic view of the tiles with their identifiers.

The red dots in the corners of tiles indicate the positions of the sensors. The walls are shown in black. The tiles under the furniture in the sleeping room (top left room), under the toilet in the bathroom (top middle room), and under the kitchen furniture (top right room) have been left unequipped with sensors, and are shown here in grey.

III. MATERIALS AND METHODS

The data generated by the floor can be passed to a data processing software either as a recorded .txt file, or through a ROS publisher/listener interface by publishing it on the corresponding topic. The software treats the data packets sent regularly by the tiles every 20 milliseconds. Data is finally converted to .mat file, which is readable by Matlab.

A. Floor Capabilities

The pressure sensors (strain gauge load cells) measure the load forces exerted on the floor that can be used to determine, for example, where you are stood on the floors surface and the way you are stood. In addition, we will proceed in this paper to determine the posture of the monitored person (walking, sitting, standing, falling, etc.) by combining the pressure amount, the pressure duration on a tile, and the tiles proximity (the floor surface exerted by load forces) using a relatively simple algorithm. As a consequence, tiles can be exploited to extend agents' perceptions and communications (of human or robot) to offer a quick assistance after fall detection alert [32,33]. One

can see an example of format of the real data sent by a tile as in the table 1.

TABLE 1
FORMAT OF DATA PACKAGE CONTAINING PRESSURE MEASUREMENTS

Information				Load Data			
Sensor type	Tile ID	msg ID	Timestamp	S1	S2	S3	S4
Gauge	45	4648	1.455E+12	1920	919	2174	212

Tile is equipped by an embedded computational components permit to measure pressure, as well as memorize, compute and transmit data. The sensors are queried for measurement data with a frequency around 50 Hz.

The tiles have also embedded 3-axis accelerometers that detect and localize hard human falls. However, the problem persists for soft falls that are difficult to detect using accelerometers. One can see the format of the accelerometer data sent by a tile in the table 2.

TABLE 2
FORMAT OF DATA PACKAGE CONTAINING ACCELEROMETER MEASUREMENTS

Information				Accelerometer Data		
Sensor type	Tile ID	msg ID	Timestamp	X	Y	Z
Lsm	18	15786	1.45562E+12	-46	18	-1072

Besides being equipped with a set of sensors, each tile also has an on-board processing unit, as well as a wireless and wired connection, which also provides electric power. For general purposes, the tiles can also communicate with other entities to allow them, for instance, to provide navigation aid for mobile robots. Further, each tile can be used as a weighing scale. Moreover, the tiles provide feedback about the weight placed on them, using the LEDs and a color code. They can be also used as Tetris game and a visual guidance for elders at night. Consequently, we can implement a large number of functionalities, including weight measurement [7], fall detection, extraction and tracking of footsteps, recognition of daily activities, activities' transition, localization and tracking people, as well as fall detection using the force sensors.

We focus on detecting all types of fall, and differentiating between the main Activities of Daily Living (ADL) (walking, sitting, standing, and falling) equally the transitions among them. The following diagram shows the main ADL states and the transitions among them "Fig. 3".

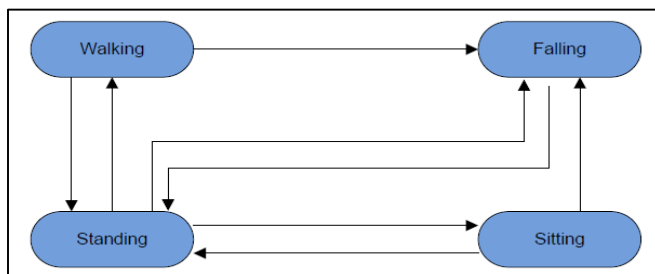


Fig. 3. State diagram of the main ADL with the transitions among them.

B. Characteristics of all postures

The work contribution starts by identifying the characteristics of different kinds of posture based on the load forces exerted on

the floor and other information (the tiles proximities and the duration). Regarding the scenarios of all occurrences, we observe the characteristics of each activity and we can note the following:

- *Walking*: the load forces exerted on the floor are high and move on neighbor tiles within a short time period.
- *Standing*: the load forces exerted on the floor are high and they are fixed on the same tiles for a time period.
- *Sitting*: the load forces exerted on the floor are split among less than 2 linear tiles and they are fixed on the same tiles for a time period.
- *Falling*: the load forces exerted on the floor are split among at least 3 linear tiles and they are fixed on the same tiles for a time period.
- *Lying down*: this posture has the same characteristics of the falling one regarding exerted the load forces.

We also take in consideration the situation where the person is going outside or existing in an area that is unequipped with sensors (i.e. taking a bath or a shower in the bathroom), and the case where there is a sensor defect. In such cases, there is no load force exerted on tiles, and they are denoted as "Unknown".

Note: The number of linear tiles is the number of tiles in a same line or in a same column.

IV. DATA EXTRACTED FROM THE INRIA-NANCY SMART TILES

Six persons were participated to make 10 different scenarios repeated 5 times. These scenarios contain the different activities: walking, sitting, standing, and falling. Activities were made with different ways like soft and hard falls, fast and slow walking, falling from sitting and falling from walking, etc. From the signals generated by the force sensors and the accelerometers under the tiles, a set of parameters has been extracted to build together our database. Furthermore, artificial intelligence techniques can be profitably used on it for elder fall detection purposes.

A. Extracting data from the force sensors

Experiments have been done in the living room having the tiles ID from 1 to 55. From the gauge database tables, we obtain 4 matrices (5x11) where their values correspond to the load forces exerted on the 4 force sensors of each tile. Also we calculate another matrix (5x11) where their values correspond to the mean load forces exerted on the 4 force sensors of each tile. The load forces were calculated after being normalized and filtered from noise, and they were reset to zero where no person exists on the tiles.

1) Algorithm

The algorithm searches the maximal value of the mean load forced exerted on each tile and then compare this value with two predefined thresholds. The first (threshold1) can distinguish the walking/standing postures from the sitting/falling/lying-down/unknown ones. Then the proposed method differentiates between walking and standing up postures regarding the duration of load force exerted on a tile using its identifier and comparing it with the previous one. On

the other hand, the second threshold (threshold2) distinguishes between the unknown posture and the sitting/falling/lying down. Then, the sitting was differentiated from falling/lying down based on the number of linear tiles exerted by load forces (tiles' proximity). See the characteristics of all postures defined in the section III.B.

Note that the threshold1 and the threshold2 are estimated to 3/4 and 1/4, respectively, of the maximum value of the load forces exerted on tiles. Since, we found when observing the forces exerted on the tiles that the maximal values were obtained when the person is walking or standing up, so we choose the threshold 75%. Whereas these values are lower in other postures because the load force is distributed on different tiles, so we choose the threshold 25%.

The detailed flow chart of the algorithm is shown in “Fig. 4”.

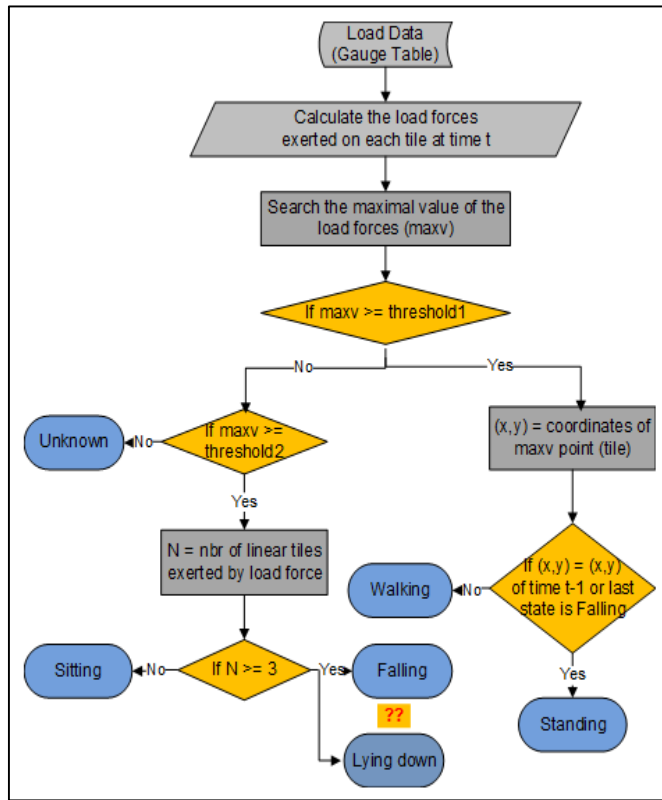


Fig. 4. Flow chart diagram of the algorithm.

This system has shown high accuracy to locate elder and estimate the main Activities Daily Living (ADL). Unfortunately, the pressure sensor is insensitive to lying down posture, and it generate a false alarm of falling since more than three linear tiles are exerted by load forces for the both postures. An alarm from the force sensor will signify that someone was falling, even the person was lying down. Therefore, we propose to fuse force sensors and accelerometers to differentiate between falling and lying down postures.

B. Extracting data from the accelerometer

The accelerometer is used to enforce the differentiation between the falling and the lying down posture. Firstly, the database tables are fragmented into fall and non-fall fragments to obtain two different classes, and then we proceed to do the windowing of each signal in each class. From each window, we

extract some useful parameters and then select the most pertinent of them in order to be used to differentiate between the two states: falling and non-falling. The accelerometer data processing is shown in “Fig. 5”.

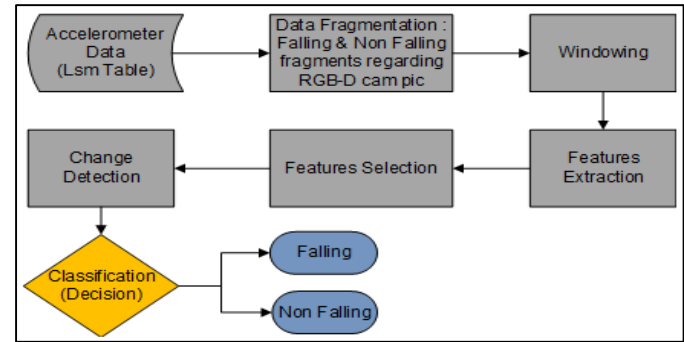


Fig. 5. Accelerometer data processing.

1) *Data fragmentation*: The start and the end of each fall action have been annotated based on the pictures and the videos taken from RGB-D cameras fixed on the wall. Similarly, for each normal activity case, suitable start and end times have been chosen so that the delimited time span contains activities as similar as possible to falls, i.e. with highly varying signals.

2) *Windowing*: We divide each signal into smaller time segments called windows of fixed length and with a fixed overlap. Good values for both window size and overlap have been found through preliminary experiments as 50 samples and 25 samples, which correspond to 1 second and 0.5 second respectively.

3) *Features extraction*: Features or parameters extraction is the process by which relevant characteristics or attributes are identified from the collected data. Features have to be cautiously picked in order to get a more descriptive and usually smaller output dataset. From each window, we have extracted all useful linear and non-linear parameters.

a) Linear parameters

i) Parameters related to power spectral density: Several frequency parameters are extracted from the power spectral density (PSD). In our work, we use the Welch Periodogram method to calculate the power spectral density. Four frequency parameters are extracted from this PSD: mean frequency MPF [34], peak frequency PF [35,36], deciles D1...D9 [37] that contain the median frequency D5 [36,367,38]. Deciles correspond to frequencies D1...D9, that divide the power spectral density into parts containing 10% of total energy.

$$\int_{D_{p-1}}^{D_p} S_x(f)df = 0.1 \int_0^{f_{max}} S_x(f)df \quad (1)$$

ii) Parameters extracted from wavelet packet decomposition: The decomposition of signals into orthonormal bases (discrete wavelet transform) is based on the theory of multi-resolution analysis. This theory proves that we can analyze a signal by decomposing it into approximation and detail coefficients [39]. The approximations correspond to smoothed versions of the low-pass filtered signal, in which changes are gradually rubberized or attenuated. The details contain only the information of high frequencies or discontinuities. After the decomposition of each window signal

into detail coefficients, we calculate the variances on these details and then used these variances for classification [40].

b) Non-linear parameters

i) Time reversibility: A time series is reversible if the probabilistic properties are unchanging with respect to time reversal. The time irreversibility is a good indication of nonlinearity. The time reversibility (TR) characteristic of a signal x is then calculated as following (for more details see [36]):

$$Tr(\tau) = \frac{1}{N-\tau} \sum_{d=\tau+1}^N (x(d) - x(d-\tau))^3 \quad (2)$$

where N is the signal length and τ is the time delay.

ii) Lyapunov Exponent: The Lyapunov Exponent (LE) studies the stability and the sensibility on initial conditions of the system. It measures the rate of trajectory separation between adjacent tracks in phase space [41,42]. In our study, we have used the “(2)” to calculate LE described in [41]:

$$\lambda = \lim_{t \rightarrow \infty} \lim_{\|\Delta_{d0}\| \rightarrow 0} \left(\frac{1}{t} \right) \log(\|\Delta_{dt}\| / \|\Delta_{d0}\|) \quad (3)$$

where $\|\Delta_{d0}\|$ represents the Euclidean distance between two states of the system to an arbitrary time t_0 , and $\|\Delta_{dt}\|$ corresponds to the Euclidean distance between the two states of the system at a time later t .

iii) Sample Entropy: To identify the regularity of signals, the Sample Entropy (SE) is used. In our work, we have used the Sample Entropy described in [38]. They indicated that a least predictable time series have higher sample entropy. Considering a time series x which represents a signal of length N and patterns aj ($0, \dots, m-1$) of length m , with $m < N$, and $aj(i) = x(i+j)$; ($i = 0, \dots, m-1; j = 0, \dots, N-m$). The time series x in a time $t = ts$, $x(ts, \dots, ts+m-1)$ as a match for a given pattern aj , if $|x(ts+i) - aj(i)| \leq r$ for each $0 \leq i < m$. Sample Entropy is then computed as follows:

$$SE_{m,r}(x) = \begin{cases} -\log\left(\frac{C_m}{C_{(m-1)}}\right): C_m \neq 0 \wedge C_{(m-1)} \neq 0 \\ -\log\left(\frac{N-m}{N-m-1}\right): C_m = 0 \vee C_{(m-1)} = 0 \end{cases} \quad (4)$$

where the four parameters N , m , r and C_m represent, respectively, the length of the time series, the length of sequences to be compared, the tolerance for accepting, and the number of pattern matches (within a margin for r) that is constructed for each m . The value of m is determined by the method of the false nearest neighbors (FNN); the value of r equals 0.2 according to the literature [38].

4) Features selection: The number of parameters in a dataset has a direct impact on the dataset's descriptive power. The more features a dataset has, the more expressive it is. However, finding meaningful relationships among the instances and the class can be more difficult because the feature space grows exponentially with the number of features. On the other hand, by selecting only the features that best describe the input data and discarding redundant features, the size of the dataset is reduced and it is very helpful in improving computational speed and prediction accuracy [43]. Several feature selection methods have existed in literature (SFS, ReliefF, F-score, etc.), and each

one has strengths and weaknesses. In this paper, we introduce and theoretically examine a new method, **HCM** (Histogram Comparison Method) and a combination of this method with another well-known method, **HCM-SFS** (HCM combined to SFS (Sequential Forward Selection)).

a) SFS: Is the first method proposed in 1963 by Marill and Green for features selection. SFS starts with an empty data set and proceeds by expanding the data set with the feature, of which addition to the data set boosts the wrapped model performance most. The algorithm adds features in such manner recursively until a stopping criteria is met [44].

b) ReliefF: Is an extension of the basic Relief algorithm which works only with the two-class problems to the multiple-class problems. It aims to estimate the quality of features according to how well their values separate the instances depending on their distance in the problem space. Given a randomly selected instances, the algorithm searches for the k nearest neighbors from the same class and k nearest neighbors from each of the other possible classes. Based on which class do the neighbors belong to, the algorithm updates the feature quality information by increasing its value if the feature separates instances with different classes well and by decreasing its value in the opposite scenario. The process of random instance selection is repeated for several times, where the number of iterations is pre-chosen by the user [45].

c) F-score: Is a simple and effective criterion to measure the discrimination of each feature. Based on statistic characteristics, it is independent of the classifiers. F-score is given by:

$$F(i) = \frac{(\bar{X}_i^{(+)} - \bar{X}_i)^2 + (\bar{X}_i^{(-)} - \bar{X}_i)^2}{\frac{1}{n_+ - 1} \sum_{k=1}^{n_+} (X_{k,i}^{(+)} - \bar{X}_i^{(+)})^2 + \frac{1}{n_- - 1} \sum_{k=1}^{n_-} (X_{k,i}^{(-)} - \bar{X}_i^{(-)})^2} \quad (5)$$

where $\bar{X}_i^{(+)}$, $\bar{X}_i^{(-)}$ and $\bar{X}_i$ are the averages of the i th feature of the positive, negative and whole datasets; n_+ and n_- are the number of positive and negative instances respectively; and $X_{k,i}^{(+)}$ and $X_{k,i}^{(-)}$ are the i th feature of the k th positive instance and the i th feature of the k th negative instance. A larger $F(i)$ indicates that the feature is more discriminative. A known deficiency of F-score is that it considers each feature separately and therefore cannot reveal mutual information between features. However, F-score is simple and generally quite effective [46].

d) HCM: Is a novel filter model which calculates the discriminative ability of each feature. That is to say, features with higher score have better separation ability in classification problems. It is a simple and effective criteria to measure the discrimination between a feature and the label, and it is also independent of the classifiers. The algorithm of HCM is defined as the following: We assume two-class classification problems. An instance is represented by a vector composed of n parameter values. For each parameter P_i , we calculate the Score value by going from the maximum value to the minimum value of the parameter with a fixed threshold, and for each step, we count the difference between the numbers of instances belong the two classes that are above the threshold (i.e. $d(P_i)$). Finally, we accumulate the differences denoted Score value (S_i) (i.e. $S_i =$

$\sum d(P_i)$. The parameter that has the higher Score value is the most discriminant. HCM is computed as the follows:

$$S_i = \sum \sum_{threshold}^{+\infty} |N^+ - N^-| \quad (6)$$

where N^+ and N^- are respectively the number of instances belong the first and the second class, that are above the threshold [25].

e) *HCM-SFS*: HCM can only examine the discriminative ability of each individual feature. Hence, features with low scores will be disregarded, even if they are complementary to the top features and might be very useful when they have combined with others parameters than were considered each one lonely. In the other hand, SFS yields the discriminative capacity of multiple features but it also suffers from the nesting effect; a parameter added in a given step cannot be removed during the next step. Hence, we introduce the HCM-SFS method which combines the two previous methods HCM and SFS. Relevant features selected by HCM are afresh passed to SFS for pick out the most pertinent features group of them. Consequently, we get rid from the problem of choosing features individually presented in HCM and also from nesting problem of SFS.

5) *Signal change detection*: is the ability to detect that a change occurred on the characteristics of a signal and making some inference about the actual time of change. Most detection algorithms are based on the theory of statistical detection and hypothesis testing [47]. In our application, we developed a simple algorithm to detect the characteristic change of a signal and notice that a person is falling down after he/she was walking, sitting, or standing. Identifying the likely time that a fall may have occurred is called change point identification.

V. EXPERIMENTS

We evaluate our approach by running experiments with humans performing the five preceded activities. Results show that our proposed system can differentiate among the different ADL postures and can detect all types of falls with a good accuracy.

A. Pressure sensors experiments

To illustrate this procedure, a program was developed using Matlab which shows the different information and decisions depending on the video sequence (several sequences have been realized but only one is presented here). Six postures are shown in the six figures below. The unknown posture mean there is no person on the floor.

Each figure contains four pictures that represent:

(a): the video snapshot taken for a subject

(b): the exerted force on a tile for a period of time

(c): the cartography of the apartment represented as image. The white cells are cells that are identified as occupied, while the dark cells have identified as free (due to the absence of pressure on the tile).

(d): the posture estimated by our system with the trajectory followed by the person.

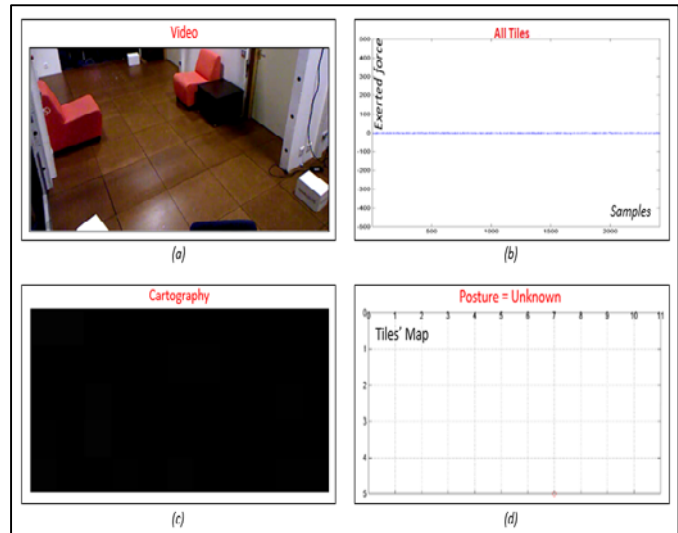


Fig. 6. Unknown posture.

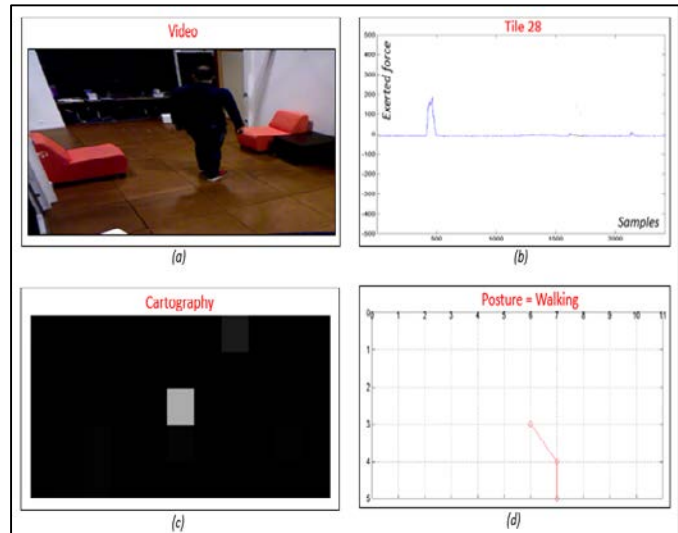


Fig. 7. Walking posture.

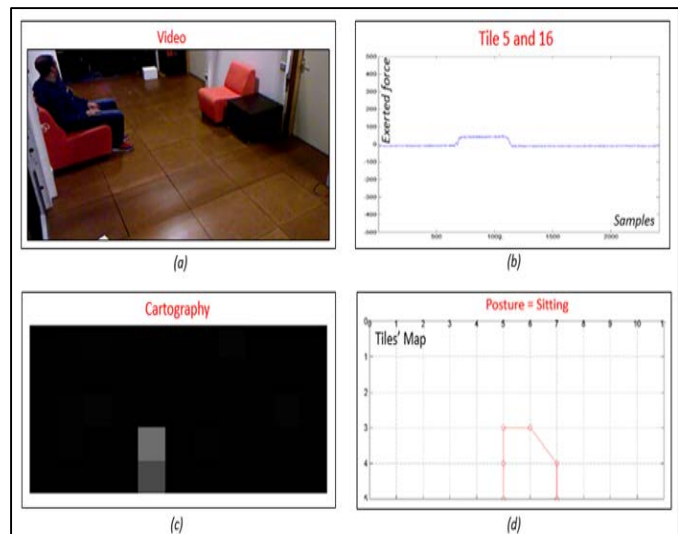


Fig. 8. Sitting posture.

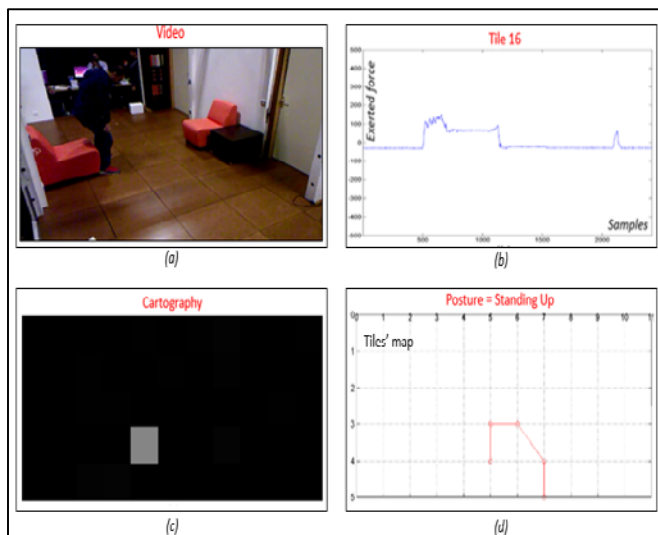


Fig. 9. Standing up posture.

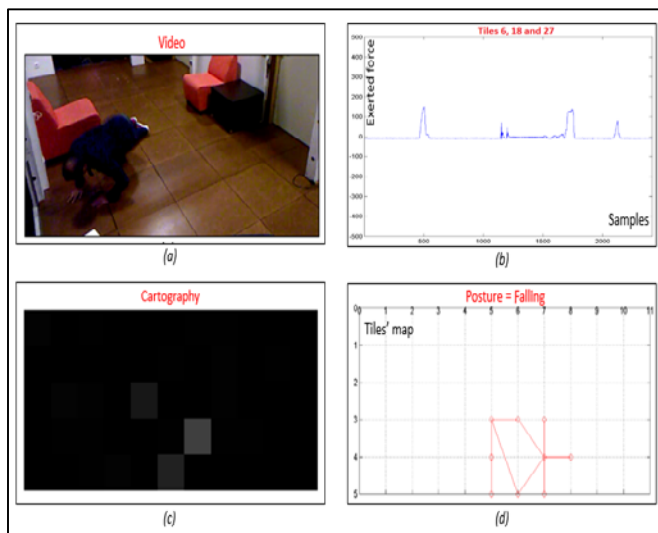


Fig. 10. Falling posture.

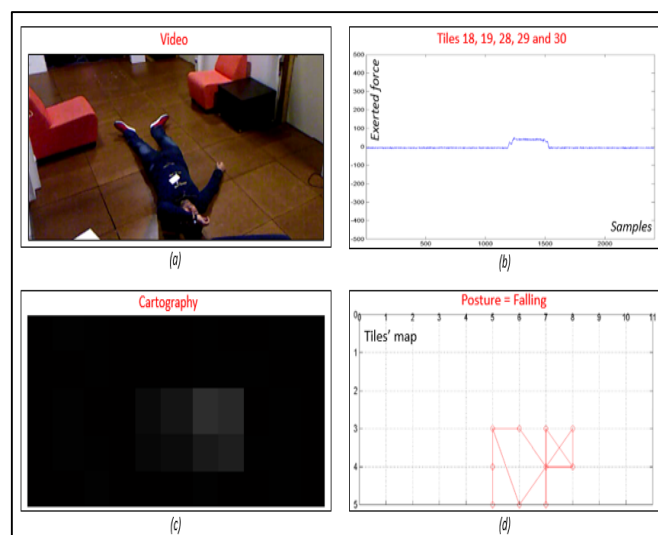


Fig. 11. Lying down posture.

One can see that by using simple pressure sensors all estimated postures are correct comparing with the video snapshots, unless the lying down posture is estimated as falling. To solve this issue, we propose to fuse between the force sensors measurement and the accelerometer decisions.

B. Accelerometer experiments

Firstly, the proposed method is tested to select the discriminative parameters on synthetic data and evaluate its efficiency compared to well-known methods in literature. After that, we will show the results of parameters' extraction and selection, and then the change detection on real data. Note, that the accelerometer is used to enforce the differentiation between the falling and the lying down posture.

1) Features selection Experiments on synthetic data

A synthetic database is generated with a Gaussian noise, composed of 200 observations that are divided into two classes each of 30 parameters around the value (0) for the first class, and around the value $(0 \pm \Delta)$ for the second class, where " Δ " is the difference between the two classes. Our aim is to test the efficiency of HCM regarding its accuracy to select relevant parameters and its time delay. Our method shows a good accuracy in relevant parameters selection. For a difference value between two classes (Δ) greater than 0.4, the error rate is around 1% which is clearly lower than ReliefF (30%) and F-score (12%) as shown in the "Fig. 12".

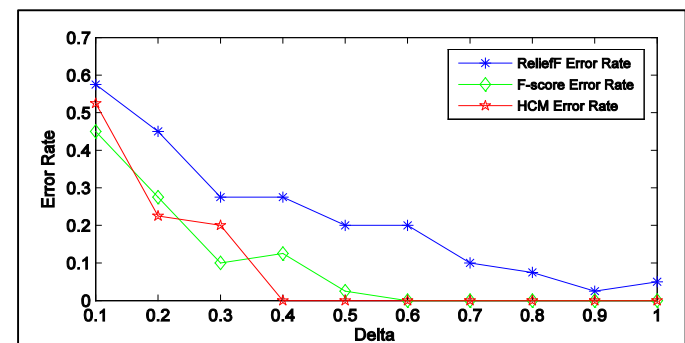


Fig. 12. Comparison of ReliefF, F-score, and HCM Error Rates.

We also have been noticed that the time latency of HCM (0.03 s) is less than the ones of ReliefF (0.05 s) and F-score (0.045 s). This characteristic allows this method to be used in some applications where the time is critical. These results are taken from a machine Intel® Core™ i5-3317U CPU @ 1.70 GHz (4 CPUs), ~ 1.7 GHz with 4 GB RAM.

Since HCM identifies only the discriminative ability of individual feature, and SFS suffers from the nesting effect, we integrated the HCM-SFS method that determines the distinctive capacity of multiple features with better accuracy than SFS, especially for a low difference classes value ($\Delta < 0.4$). For example, the error rate is decreasing from 72% to 28% for a ($\Delta = 0.1$) "Fig. 13".

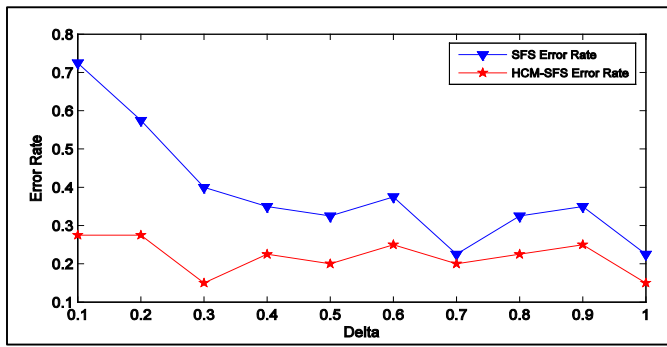


Fig. 13. Comparison of SFS and HCM-SFS Error Rates.

2) Experiments on real data

We try here to discriminate between falling and non-falling states using the accelerometer signals. We took a signal when a person is walking, standing, sitting, falling, and lying down. After that, we extract some useful parameters from the original signals that can be used in our system in order to improve the performance. Some extracted parameters are shown in the “Fig. 14” in function of the number of windows. This signal is normalized before being drawn.

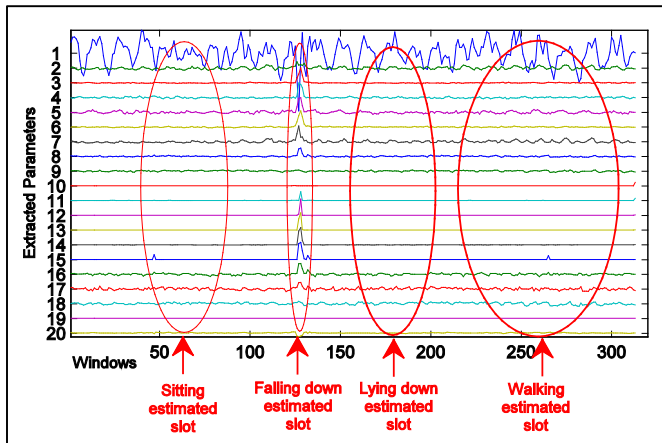


Fig. 14. Some Extracted Parameters (1: TR, 2: DFA, 3-7: Wavelet Packet decomposition parameters, 8: MPF, 9-18 Decil, 19: SE, and 20: LE).

It is clear that most of extracted parameters have a high peak in case of fall. Thus, we took one of the most relevant parameters and then we developed a simple algorithm to detect the change in falling cases. “Fig. 15” shows the signal in (a), and the fall detection with a good accuracy in (b).

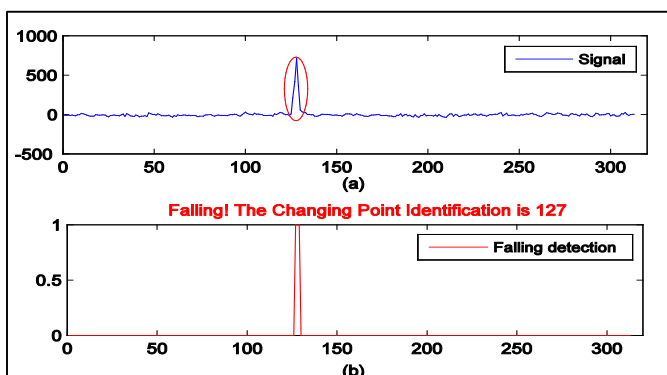


Fig. 15. Signal change detection when a fall occurs.

Therefore, using a simple accelerometer data, we can easily differentiate between falling and non-falling states.

C. Experiments of fusion between the force sensors and the accelerometer decisions

We show that the force sensors signals can be used to differentiate among walking, standing, sitting, unknown states, but they confuse between falling and lying down postures. In addition, we show also that accelerometer signals can easily differentiate between the falling and non-falling states. Hence, by using the data fusion of the force and accelerometer sensors, we can differentiate among walking, standing, sitting, unknown, falling and lying down states. The complete process flow chart of the system is shown in the “Fig. 16”.

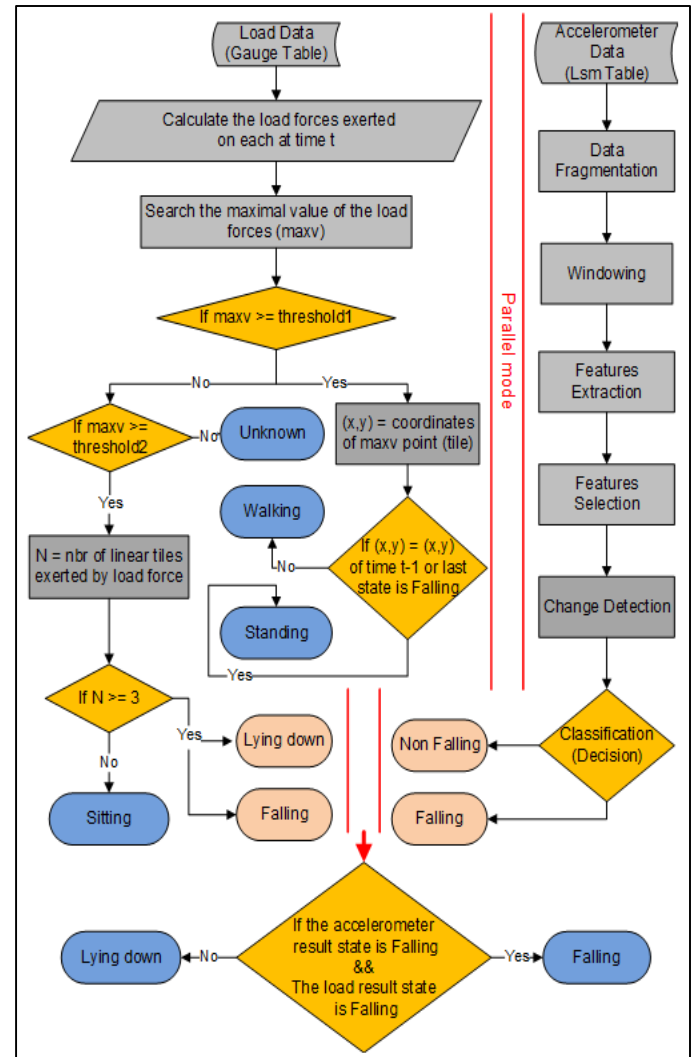


Fig. 16. Flow chart diagram of the system.

To end up, the proposed system permit finally to locate and track elder, recognize their activities, and detect fall cases.

VI. RESULTS AND DISCUSSIONS

Ten different sequences with six different subjects were tested to show the accuracy and the efficiency of the proposed system. “Table 3” shows the sensitivity results for different ADL postures.

TABLE 3
SENSITIVITY FOR DIFFERENT ADL POSTURES

Real postures	Estimated postures	Sensitivity
60 Unknown	60 Unknown	100%
150 Walking	148 Walking + 2 Standing Up	98.7%
65 Standing Up	60 Standing Up + 3 sitting + 2 Unknown	92.3%
42 Sitting	40 Sitting + 2 Unknown	95.2%
34 Falling	32 Falling + 1 Sitting + 1 Walking	94.1%
20 Lying down	18 Lying down + 1 Sitting + 1 Falling	90%

The first column describes the real postures of the six subjects in the ten sequences. The second column shows the estimated postures of the proposed methods. The third column shows the sensitivity or the True Positive Ratio (TPR = TP / (TP+FN)) of the proposed system, where TP and FN are respectively the numbers of valid and invalid estimated postures, 100% indicating that all real postures were truly estimated.

Thus, one can see that the proposed methods are accurate (the system sensitivity is greater than 90% for all estimated postures). To prove this accuracy, a supplementary multimedia file has been attached to this paper showing the precision of the postures estimation. In addition, for a sequence example (walking – sitting – falling - standing up - walking), the mass center localization is drawn in the “Fig. 17” to validate the estimation.

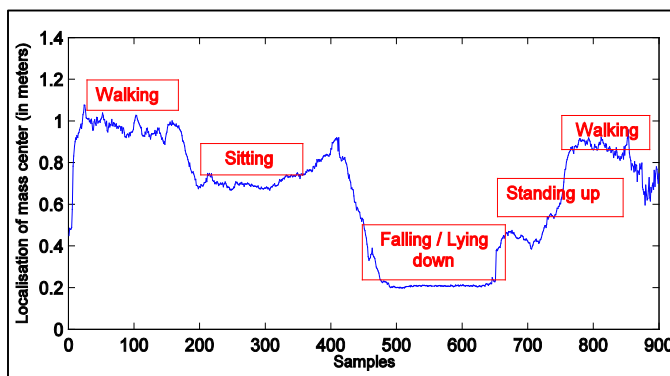


Fig. 17. Localization of the mass center of a person.

Furthermore, by extracting some meaningful features from the collected data and selecting only the pertinent ones, the size of the dataset is reduced and becomes more efficient. The efficiency appears by improving the computational speed and the prediction accuracy of the proposed system.

On the other hand, most of existing fall detection systems use wearable sensors mounted on the person or a camera-based systems that are uncomfortable to the elderly. Moreover, most of prevailing sensing floors are used to track, locate and detect human fall cases. Whereas, the idea behind this work is to provide a system that not only tracks, locates and detects elders' falls, but also recognizes their activities (walking, standing, sitting, lying down, falling, and the transitions between them), using a set of non-intrusive sensors that grants privacy and comfort to the elders.

Indeed, the large-scale deployments of the proposed system require a large number of sensors and are expensive. However, due to the recent rapid advancement in information and

communication technologies, sensors will be cheaper in the near future to reduce the cost of the system installation.

Finally, as a perspective, we will try to use an automated thresholding method that calculates optimal and adaptive thresholds to use them in the force sensor algorithm.

VII. CONCLUSION

This paper proposes a fine elder tracking, activities recognition, and fall detection system using a set of non-intrusive sensors. The proposed system uses the fusion between pressure sensors and accelerometers hidden under the smart tiles.

First, we presented the aging population problem and the advances in medicine and public health services to track and monitor activities of elderly people at home in order to assist their independence.

Second, we showed the proposed algorithm and results to distinguish different postures using the signals generated by force sensors of the INRIA-Nancy smart tiles platform. We deduced that force sensors cannot differentiate between falling and lying down postures because they have the main characteristics in term of the load exerted on tiles, tiles proximities, and the duration.

Third, we proceeded to the accelerometer signals to differentiate between the falling and lying down postures. We had extracted some useful parameters and then selected the pertinent one to detect falls.

Fourth, we showed the flowchart of the fusion between the force sensors and the accelerometers that finally permitted the ability to locate and track the elders, recognize their activities, and detect fall cases.

To end up, our challenge is to offer a usable system with maximum privacy within a reliable, efficient and affordable framework.

REFERENCES

- [1] Z. Zhou, et al., "A real-time system for in-home activity monitoring of elders." Engineering in Medicine and Biology Society, 2009. EMBC 2009. Annual International Conference of the IEEE. IEEE, 2009.
- [2] "Aging, can we stop the clock?", Wellcome Trust Report (<http://www.wellcome.ac.uk/>).
- [3] J.A. Stevens, "Falls among older adults: risk factors and prevention strategies." Journal of safety research 36.4 (2005): 409-411.
- [4] V. Spasova and I. Iliev, "Computer Vision and Wireless Sensor Networks in Ambient Assisted Living: State of the Art and Challenges," Journal of Emerging Trends in Computing and Information Sciences, vol. 3, no. 4, 2012, pp. 585-595.
- [5] A. Bamis, et al., "Towards precision monitoring of elders for providing assistive services." Proceedings of the 1st international conference on Pervasive Technologies Related to Assistive Environments. ACM, 2008.
- [6] N. Pepin, O. Simonin, and F. Charpillat, "Intelligent Tiles-Putting Situated Multi-Agents Models in Real World," ICAART. 2009.
- [7] M. Andries, O. Simonin, and F. Charpillat, "Localisation of humans and objects interacting on load-sensing floors," IEEE Sensors Journal 16, no. 4 (2016): 1026-1037.
- [8] X. Yu, "Approaches and Principles of Fall Detection for Elderly and Patient", 10th IEEE Intl. Conf. on e-Health Networking, Applications and Service (HEALTHCOM 2008), July 2008.

- [9] M.D. Addlesee, A. Jones, F. Livesey, and F. Samaria. "The ORL active floor". In: *Personal Communications, IEEE 4.5* (1997), pp. 35–41. issn: 1070-9916. doi: 10.1109/98.626980.
- [10] J. Paradiso, et al. "The magic carpet: physical sensing for immersive environments." *CHI'97 Extended Abstracts on Human Factors in Computing Systems. ACM*, 1997.
- [11] R. J. Orr and G. D. Abowd. "The smart floor: a mechanism for natural user identification and tracking". In: *CHI'00 Extended Abstracts on Human Factors in Computing Systems*, pp. 275-276. ACM, 2000.
- [12] S. Helal, et al. "The gator tech smart house: A programmable pervasive space." *Computer* 38.3 (2005): 50-60.
- [13] CD. Kidd, et al. "The aware home: A living laboratory for ubiquitous computing research." *Cooperative buildings. Integrating information, organizations, and architecture. Springer Berlin Heidelberg*, 1999. 191-198.
- [14] T.S.T. MORI, "Robotic room: Its concept and realization." *Intelligent Autonomous Systems: IAS-5* (1998): 415.
- [15] S. Tomomasa, T. Harada, and T. Mori. "Environment-type robot system" *RoboticRoom*" featured by behavior media, behavior contents, and behavior adaptation." *Mechatronics, IEEE/ASME Transactions on* 9.3 (2004): 529-534.
- [16] D. De, et al. "ActiSen: Activity-aware sensor network in smart environments." *Pervasive and Mobile Computing* 8.5 (2012): 730-750.
- [17] K. Nakajima, et al. "Footprint-based personal recognition." *Biomedical Engineering, IEEE Transactions on* 47.11 (2000): 1534-1537.
- [18] M. Razeghi, and M.E. Batt. "Foot type classification: a critical review of current methods." *Gait & posture* 15.3 (2002): 282-291.
- [19] T. Kuragano, A. Yamaguchi, and S. Furukawa. "A method to measure foot print similarity for gait analysis." *Computational Intelligence for Modelling, Control and Automation, 2005 and International Conference on Intelligent Agents, Web Technologies and Internet Commerce, International Conference on. Vol. 2. IEEE*, 2005.
- [20] A. Braun, H. Heggen, and R. Wichert. "CapFloor—a flexible capacitive indoor localization system." *Evaluating AAL Systems Through Competitive Benchmarking. Indoor Localization and Tracking. Springer Berlin Heidelberg*, 2011. 26-35.
- [21] A. Ropponen, H. Rimminen, and R. Sepponen. "Robust system for indoor localisation and identification for the health care environment." *Wireless Personal Communications* 59.1 (2011): 57-71.
- [22] H. Rimminen, J. Lindström, and R. Sepponen. "Positioning accuracy and multi-target separation with a human tracking system using near field imaging." *International Journal on Smart Sensing and Intelligent Systems* 2.1 (2009): 156-175.
- [23] H. Rimminen, et al. "Detection of falls among the elderly by a floor sensor using the electric near field." *IEEE transactions on information technology in biomedicine: a publication of the IEEE Engineering in Medicine and Biology Society* 14.6 (2010): 1475-1476.
- [24] H. M. Schepers, et al. "Ambulatory estimation of foot placement during walking using inertial sensors." *Journal of biomechanics* 43.16 (2010): 3138-3143.
- [25] T. Liu, Y. Inoue, and. Shibata. "A wearable ground reaction force sensor system and its application to the measurement of extrinsic gait variability." *Sensors* 10, no. 11 (2010): 10240-10255.
- [26] T. Liu, Y. Inoue, K. Shibata, K. Shiojima, and M. M. Han. "Triaxial joint moment estimation using a wearable three-dimensional gait analysis system." *Measurement* 47 (2014): 125-129.
- [27] D. Li, X. Zhu, and J. Zhe. "An Inductive Sensor for Real-Time Measurement of Plantar Normal and Shear Forces Distribution." *IEEE Transactions on Biomedical Engineering* 62, no. 5 (2015): 1316-1323.
- [28] C. Lauterbach, A. Steinhage, and A. Techmer. "Large-area wireless sensor system based on smart textiles." *International Multi-Conference on Systems, Signals & Devices*. 2012.
- [29] M. Sousa, et al. "Human tracking and identification using a sensitive floor and wearable accelerometers." *Pervasive Computing and Communications (PerCom)*, 2013 IEEE International Conference on. IEEE, 2013.
- [30] Tarkett. *FloorInMotion*. url: <http://www.floorinmotion.com/>.
- [31] M. Daher, M. El Badaoui El Najjar, A. Diab, M. Khalil, and F. Charpillat. "Towards a usable and an efficient elder fall detection system." 2015 International Conference on Advances in Biomedical Engineering (ICABME). IEEE, 2015.
- [32] M. Rio, F. Colas, M. Andries, and F. Charpillat. "Probabilistic sensor data processing for robot localization on load-sensing floors." 2016 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2016.
- [33] G. Sannino, I. De Falco, and G. De Pietro, "Automatic Extraction of an Effective Rule Set for Fall Detection for a Real-Time Mobile Monitoring System." *Developments in eSystems Engineering (DeSe)*, 2013 Sixth International Conference on. IEEE, 2013.
- [34] J. Terrien, T. Steingrimsdottir, C. Marque, and B. Karlsson, "Synchronization between EMG at different uterine locations investigated using time–frequency ridge reconstruction: comparison of pregnancy and labor contractions," *EURASIP J. Adv. Signal Process.* 2010, 1–10 (2010).
- [35] W.L. Maner, R.E. Garfield, H. Maul, G. Olson, and G. Saade, "Predicting term and preterm delivery with transabdominal uterine electromyography." *Obstet. Gynecol.* 101(6), 1254–1260 (2003).
- [36] J. Sikora, A. Matonia, R. Czabanski, K. Horoba, J. Jezewski, and T. Kupka, "Recognition of premature threatening labour symptoms from bioelectrical uterine activity signals," *Arch. Perinat. Med.* 17(2), 97–103 (2011).
- [37] C. Marque, H. Leman, M.L. Voisine, J. Gondry, and P. Naepels, "Traitement de l'électromyogramme utérin pour la caractérisation des contractions pendant la grossesse," *RBM-News* 21(9), 200–211 (1999).
- [38] G. Fele-Žorž, G. Kavšek, Ž. Novak-Antolić, and F. Jager, "A comparison of various linear and non-linear signal processing techniques to separate uterine EMG records of term and pre-term delivery groups." *Med. Biol. Eng. Comput.* 46(9), 911–922 (2008).
- [39] R.R. Coifman, and M.V. Wickerhauser, "Entropy-based algorithms for best basis selection", *IEEETrans. Inf. Theory* 38(2), 713–718 (1992)
- [40] D. Alamedine, M. Khalil, and C. Marque, "Comparison of different EHG feature selection methods for the detection of preterm labor." *Computational and mathematical methods in medicine* 2013 (2013).
- [41] A. Diab, M. Hassan, C. Marque, and B. Karlsson, "Quantitative performance analysis of four methods of evaluating signal nonlinearity: application to uterine EMG signals." Presented at the 34th Annual International IEEE EMBS Conference. San Diego, USA (2012).
- [42] T. Ivancevic, L. Jain, J. Pattison, A. Hariz, et al, "Preterm birth analysis using nonlinear methods." *Recent Patents Biomed. Eng.* 1(3), 160–170 (2008).
- [43] H.H. Hsu, C.W. Hsieh, and M.D. Lu, "Hybrid feature selection by combining filters and wrappers," *Expert Systems with Applications* 38.7 (2011): 8144-8150.
- [44] G.H. John, R. Kohavi and K. Pfleger, "Irrelevant Features and the Subset Selection Problem," *Proceedings of the Eleventh International Conference on Machine Learning*, 1994, 121–129.
- [45] I. Kononenko, "Estimating Attributes: Analysis and Extensions of RELIEF," *European Conference on Machine Learning*, 1994, 171–182.
- [46] Y. W. Chang, and C. J. Lin. "Feature ranking using linear SVM." *WCCI Causation and Prediction Challenge*. 2008.
- [47] M. Basseville, and I. V. Nikiforov. *Detection of abrupt changes: theory and application*. Vol. 104. Englewood Cliffs: Prentice Hall, 1993.



Mohamad Daher received the teaching diploma degree in applied mathematics, option informatics from Lebanese university, Tripoli, Lebanon, in 2002, and the Master's degree in computer science, option networking and communications from the University of Balamand, Al Kurah-Lebanon, in 2004. He is currently pursuing the Ph.D. degree in signal processing and control systems at Lille University - Science and Technology, CRISTAL laboratory, Lille, France.

He joined the Université de Technologie et de Sciences Appliquées Libano-Française (ULF) in 2005 as instructor and

also the Lebanese University in 2011. Since 2010, he is an academic advisor at the ULF.



Ahmad Diab received the degree in Biomedical Engineer from the Islamic University of Lebanon, Khalde, Lebanon, in 2010, and the M.Sc. degree in Medical and Industrial Processing and System from the Lebanese University, Tripoli, Lebanon, in 2011. Also he received his Ph.D. degree from the University of Technology of Compiègne, Compiègne, France and Reykjavik University, Reykjavik, Iceland in 2014.

He is currently an associate Professor and Research Engineer at Azm center for research in biotechnology and its application, Lebanese University.



Maan El Badaoui El Najjar received the computer engineering degree from ESISAR – INP Grenoble- France in 1999. He graduated from ENSIEG – INP Grenoble in 2000 (Master of Science degree in Control System). He received the Ph.D. degree in Perception and Control Systems from the University of Technology of Compiègne in 2003. In November 2011, he obtained the Habilitation à Diriger des Recherches from the University of Lille.

He joined the University of Lille in 2005 as a permanent associate Professor. Since 2014, he is Full Professor at the University of Lille. He is also with the CRISAL Laboratory UMR 9189, a joint research unit between CNRS, University of Lille and Ecole Centrale de Lille. He is also the head of the DiCOT Team “Diagnostic, Control and Observation for fault Tolerant Systems” of the CRISAL Laboratory.

The research of Prof. El Badaoui El Najjar has been focused on Multi-sensors fusion, Multi-robots systems, and Fault tolerant systems.



François Charpillet received his engineering degree from École Nationale Supérieure d'Électricité et de Mécanique (ENSEM) of the National Polytechnic Institute of Lorraine (INPL) in 1982. He received the M.S. and Ph.D. degree in computer science from Henri Poincaré University, Nancy, France, in 1982 and 1985 respectively.

After a period as an engineer at Direction des Constructions Navales (DCN) in Paris, he joined the University of Lorraine and LORIA Lab as an assistant professor in 1987. In 1988, he got a research position at Inria Nancy - Grand-Est Research Center.

Prof. Charpillet is today Director of Research at Inria. From 1998 to 2014, he has been the leader of LORIA team MAIA, which was a joint research group with Inria, CNRS and University of Lorraine. This group was interested in Artificial Intelligence focusing on autonomous agents and multi-agent systems. Since 2015, he is the scientific leader of the new Inria team Larsen, focusing on Long Term Robotics, Human-robot

interaction and Active and Distributed sensing. His current research interests concern multi-robot cooperation, human-robot- environment interaction, activity recognition, active sensing and ambient Intelligence.



Mohamad Ali Khalil received an engineering degree in electrical and electricity from the Lebanese University, faculty of engineering, Tripoli, Lebanon in 1995. He received the DEA in biomedical engineering from the University of Technology of Compiègne (UTC) in France in 1996. He received his Ph.D. from the University of Technology of Troyes in France in 1999. He received the Habilitation à Diriger des Recherches from UTC in 2006.

He is currently teacher and researcher at Lebanese University, faculty of engineering. He is classified Professor in 2008. From 2009, he is director of the Azm center for research in biotechnology at the doctoral school of sciences and technology at the Lebanese university. He is the chair of the EMBS chapter in Lebanon and also the chair of ICABME international Conference in 2011, 2013 and 2015. He organized many conferences in the Lebanese university.

Prof. Khalil was received many awards: Best supervisor 2004 and 2005 from LIRA, Best innovation project from LAsER and AUF in 2013, and Best applied research project from ministry of industry 2014. His current interests are the signal and image processing problems: detection, classification, analysis, representation and modeling of non-stationary signals, with application to biomedical signals and images. Number of publications: 45 articles in international Journals, 100 participations in conferences.